ABHISHEK DASS

LabVIEW Developer | Test & Measurement | Data Acquisition Systems

Nagpur, Maharashtra, India | +91-7798927933 | abhishekdass141516@gmail.com | LinkedIn | GitHub

PROFILE

Computer Science and Business Systems with hands-on LabVIEW development experience spanning industrial automation, medical-grade hardware testing, and real-time DAQ for electric-vehicle telemetry. At HIXAA Technologies, a LabVIEW system-integration firm, helped to build **Virtual Instrument (VI)** interfaces for a metro-simulation platform and tested a medical audiometer device before client deployment. Previously used LabVIEW to design real-time monitoring and DAQ systems for a Formula Bharat racing team.

LABVIEW & TEST ENGINEERING EXPERIENCE

Engineering Intern (LabVIEW) | HIXAA Technologies Pvt. Ltd. *Jan 2026 – Jul 2026 | Nagpur*

Industrial automation & LabVIEW system-integration firm—Testing & Measurement, IIoT, Machine Vision, Automated Test Equipment, DAQ

- **Metro Simulator—**Designed and debugged interactive LabVIEW Virtual Instrument (VI) screen UIs, building intuitive controls and readouts that accurately mimic real-world metro transit controls.
- **Arphi Audiometer—**Performed pre-deployment testing and configuration of a medical-grade audiometer device, validating hardware-software integration to meet strict accuracy client requirements and reliability requirements.
- Executed client installation and setup of deployed software and authored technical setup guidelines and user documentation for the deployment.
- Contributed to architectural planning of VI screens.

Aerodynamics & Data Acquisition Lead | E-Formula AshwaRiders *Mar 2023 – Jan 2026 | Nagpur*

Formula Bharat electric racing vehicle program—AIR 8

- **Developed LabVIEW-based real-time monitoring and data-processing systems** for continuous vehicle telemetry, covering speed, thermal, and battery-performance data.
- Architected the application using the **DQMH (Delacor Queued Message Handler) framework**, applying modular, message-based module design for scalable, maintainable LabVIEW code.
- Designed sensor-to-LabVIEW acquisition pipelines interfacing embedded hardware and wireless links under real-time reliability constraints.
- Built LabVIEW dashboards for live anomaly detection and vehicle-performance analysis, enabling data-driven engineering decisions during testing and competition.
- Managed the full data lifecycle from raw sensor input through structured storage and post-session engineering analysis.

ADDITIONAL TECHNICAL EXPERIENCE

Flight Software Team Member | Estonian Student Satellite (SUTS CubeSat) *Jul 2026 – Present | Remote*

- Contribute to Core Flight Software (cFS) applications for the satellite command module, applying real-time, fault-tolerant software practices comparable to LabVIEW-based control system design.

RELEVANT PROJECT

Real-Time Vehicle Telemetry & DAQ System — *Python | C/C++ | LabVIEW | PostgreSQL | DQMH | TCP | Embedded Systems*

- Built an end-to-end LabVIEW/Python telemetry pipeline—sensor acquisition, packet transmission, parsing, processing, visualization, and data logging—with real-time monitoring of vehicle and battery parameters.

TECHNICAL SKILLS

LabVIEW & Test/Measurement: Virtual Instrument (VI) design, front panel & block diagram development, Data Acquisition (DAQ), Automated Test Equipment, sensor integration & calibration, hardware-software integration testing, real-time monitoring dashboards, DQMH

Industrial & Embedded Systems: System Integration, Arduino, ESP32, RTOS

Programming: C, C++, Python

Engineering Tools: Git, GitHub, LabVIEW

Documentation & QA: Technical setup guides, user documentation, pre-deployment/QA testing, client-site deployment

EDUCATION

B.Tech, Computer Science and Business Systems—St. Vincent Pallotti College of Engineering & Technology, Nagpur